

benefits of VoIP could be extended to customers over the legacy infrastructure itself. Rather than challenging traditional telecom, VoIP has helped to enhance it.

Zia says, "To meet the demands of the changing business environment, to attract new customers and to add to their portfolio of revenue-generating services, service providers are deploying converged voice-and-data services based on SIP. With its foundation in Internet protocols, SIP provides the ability to integrate traditional voice services with Web-based data services, including self-based provisioning, instant messaging, presence and mobility services."

Where legally permitted, SIP has also resulted in the emergence of local exchange carriers and virtual network operators who have taken a share of the VoIP market by acting as a mediator between the PSTN and a company's LAN and providing value-added services.

So what really is SIP trunking?

A 'SIP trunk' allows a company to replace traditional fixed PSTN lines with PSTN connectivity via a SIP trunking service provider on the Internet. SIP is a lightweight module used for the setup of a session between two or more parties. It works on the application layer, using a text-based interface, and is hence easy-to-use and flexible.

To use a SIP-based service, a company would need an IP-based private business exchange (PBX) or a traditional digital or analogue PBX with a SIP trunking interface. It also needs an edge device that supports SIP, which can act as an interface between the local-area network and the Internet telephone service provider (ITSP), who, in turn, connects to the PSTN.

In short, SIP trunking enables a service provider to use the PSTN as the base to provide IP-based services at the application layer. It enables enterprises to tap the benefits of VoIP without washing out their legacy communication systems.

Players across the telecom value

chain, ranging from platform/tool providers like Microsoft, and telcos like Vantage to device makers like Cisco have SIP-based solutions.

Subject to bandwidth availability, a SIP trunk can carry unlimited voice calls (though service providers are known to restrict the number of simultaneous calls to anywhere between 28 and 41 per SIP trunk), versus a traditional PSTN/PRI trunk that can accommodate only 24 calls per channel (realistically, 17.5 calls).

According to a recent Frost & Sullivan report on VoIP and SIP trunking, although businesses of all sizes can benefit from the access-line and toll savings offered by VoIP services, SMBs have emerged as an attractive niche for VoIP access services. SMBs either deploy IP trunking as an overlay service to their existing broadband access (wire line/wireless, DSL, cable, T1/E1, Ethernet, etc) or buy pure SIP/H.323 trunks, leveraging the IP trunking provider's network and IP delivery architecture.

IP trunking using the SIP protocol is prevalent in large Indian enterprises. If more local exchange carriers crop up, and if Indian SMBs also go for it, SIP could well be a factor that could contribute to the growth of VoIP in India too.

The future of VoIP in India

As the new broadband networks are built out, we could see VoIP becoming more prevalent in the Indian market.

With the recent Broadband Wireless Access (BWA) auction, we anticipate large rollouts of WiMAX networks in the country in the near future. "WiMAX networks are based on all-IP technology and will certainly run very economical VoIP. Mobile VoIP will take a long time to be introduced in the country as today's mobile networks are largely built using circuit-switched technology, and it will take a long time to convert them to end-to-end IP networks," says Sharma.

Dr Borkar feels that even after 3G wireless becomes available in India, service providers will continue to use the

legacy circuit-switched based approach, as others are doing around the world. "That does open up the possibility for 'over-the-top' operators like Skype to provide basic VoIP capabilities but they are not even close to the carrier-grade circuit-switched voice," he says.

4G is not likely to change the situation much either. As we migrate to 4G cellular networks like Long-Term Evolution (LTE) that have only a packet core, the capability of providing voice using the data network through the SIP/IMS platform may not be applicable, from the business viewpoint, for another three-four years even in advanced countries. The operators will hence continue to leverage the circuit core (e.g., of the 3G system) for transmitting voice calls for quite some time to come. One such example is the Voice over LTE via Generic Access (VoLGA).

There is considerable work needed before VoIP can meet the carrier-grade standards, due to the statistical packet nature of the packet network, and will have to rely on very complex Quality of Service (QoS) treatment along with other applications involving data and video, explains Dr Borkar, while agreeing that ultimately all multimedia services, including voice for wireline and cellular mobile units, will be provided on the packet transport with all the feature-richness. But that eventuality is a long way off, at least five years, for reasonable market penetration.

There is no reason to immediately phase out the traditional networks as it took a lot of investment to build them and they are doing well for voice communications. However, since the advantages of VoIP are quite compelling, new VoIP networks could be built in areas where traditional telephony is lacking and also where there are serious capacity constraints. Further, SIP trunking could be used to extend legacy infrastructure to benefit from VoIP. ●

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